

Remote Sensing Platforms

IRS Chapter Three

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Roadmap

Introduction and Platforms

Fixed-Wing Aircraft

Helicopters

Satellite Systems

Unmanned Aerial Systems

Tethered Balloons

Mobile Collection of Field Data

Summary

Introduction to Remote Sensing Platforms

- ▶ Remote sensing platforms: Vehicles that carry sensors to gather information about Earth's surface.
- ▶ Applications range from ecology, geology, agriculture, cities, to forest fire monitoring.
- ▶ **Two main types of platforms:**
 - ▶ **Active remote sensing:** Emits energy and detects its reflection (e.g., radar).
 - ▶ **Passive remote sensing:** Collects naturally emitted or reflected radiation (e.g., infrared, photography).

Learning Objective

Learning Objective: Understand the distinction between different remote sensing platforms and their core applications.



Remote Sensing Platforms Overview

- ▶ Platforms carry sensor systems to collect electromagnetic radiation.
- ▶ Provide **overhead perspectives** for map-like images.
- ▶ Common platforms include:
 - ▶ **Fixed-wing aircraft**
 - ▶ **Helicopters**
 - ▶ **Unmanned Aerial Systems (UAS)**
 - ▶ **Satellites**
 - ▶ **Balloons**
- ▶ Platform choice is **critical** to remote sensing missions—impacts altitude, orientation, and timing.

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Fixed-Wing Aircraft in Remote Sensing

- ▶ **Common type of platform** with long flight ranges and precise navigation.
- ▶ Supports various sensor systems and heavy loads for different missions.
- ▶ **Advantages:** Long endurance, navigation precision, flexibility in sensor payload.
- ▶ **Disadvantages:** High costs (purchase, maintenance), delays due to weather, transit expenses.

Small Fixed-Wing Aircraft

- ▶ **Cessna 172:** Small, single-engine aircraft.
 - ▶ Features a high-wing design for **oblique photography**.
 - ▶ Commonly used for aerial photography, handheld photography, and real estate videos.
- ▶ **Camera pod:** Detachable pod allows for vertical photography and other sensor uses.



Midsized Fixed-Wing Aircraft

- ▶ **Midsized aircraft** (turbo-prop or jet-powered) support larger payloads and longer flights.
- ▶ Examples include:
 - ▶ **DeHavilland Twin Otter (DHC-6)** used by NOAA.
 - ▶ **Beechcraft King Air 90** with specific remote sensing modifications.
- ▶ Suitable for extended aerial survey missions.



Aerial Survey Firms Using Midsized Aircraft

- ▶ Aerial survey firms rely on midsized aircraft for various projects:
 - ▶ **Highway and building construction**
 - ▶ **Urban planning**
 - ▶ **Coastal and agricultural management**
- ▶ **Crews with expertise** handle navigation, extended flights, and instrument operations.
- ▶ Supported by **ground teams**: IT services, maintenance, flight planning, and image analysis.
- ▶ **Statement of Work (SOW)**: Contract specifying imagery quality, timing, and seasonal factors (e.g., leaf-off/leaf-on).

Large Fixed-Wing Aircraft (Douglas DC-8)

- ▶ **Douglas DC-8:** Four-engine, long-range jet airliner.
 - ▶ Initially used for passenger and cargo service.
 - ▶ Later reconfigured for **NASA remote sensing missions.**
- ▶ Used for missions requiring:
 - ▶ **Experimental sensors**
 - ▶ **Multisensor systems**
 - ▶ **Higher altitudes and longer missions**



NASA's DC-8 Large Aircraft

- ▶ **NASA's DC-8:** Large aircraft configured for remote sensing missions, such as testing radar systems.



SAR Antenna on NASA's DC-8

- ▶ **Radar antenna panels:** Transmit microwave pulses and receive reflected signals.
- ▶ **Apertures:** Focus the radar beams for improved resolution.
- ▶ **Synthetic Aperture Radar (SAR)** uses:
 - ▶ **Doppler effect** and **motion** for high-resolution imaging.
 - ▶ Capable of **all-weather, day-night** data collection.
- ▶ Pulse microwave energy—penetrates clouds and vegetation.



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Helicopters

- ▶ **Helicopters:** rotary-wing aircraft, allowing:
 - ▶ Ability to **hover** and fly laterally.
 - ▶ Operation in **confined areas**.
 - ▶ Maneuverability for **complex flight plans**.
- ▶ **Advantages:** Flexibility in navigating difficult terrain (e.g., forests, uneven ground).
- ▶ **Disadvantages:** High operating costs and shorter range compared to fixed-wing aircraft.



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Satellite Systems in RS

- ▶ **Artificial satellites** are critical for Earth observation, environmental data collection, and communication.
- ▶ First U.S. civil remote sensing satellite: **Landsat** (launched in 1972).
- ▶ Satellites provide:
 - ▶ **Global coverage** and a **synoptic perspective**.
 - ▶ **Repetitive, systematic coverage** for long-term monitoring.
- ▶ Used for applications such as agriculture, urban systems, and forest monitoring.



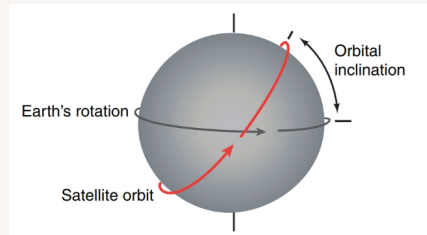
Synoptic perspective

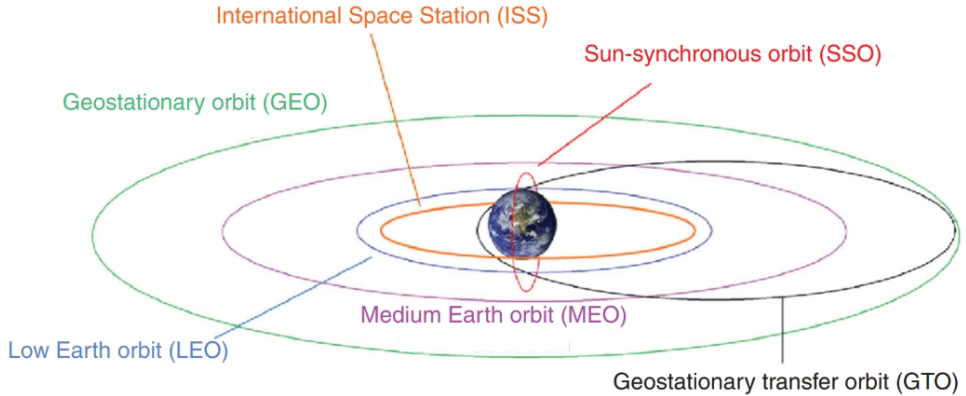
A view of a large area in one image, revealing regional patterns and relationships.

Satellite Orbits

Orbits can be describe by size elments, we mainly care about two:

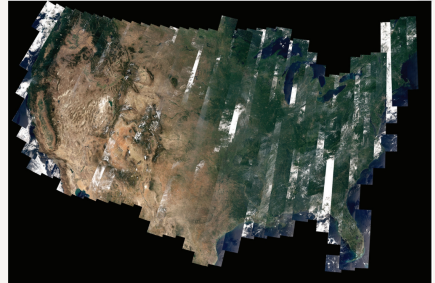
- ▶ **Inclination:** Angle of orbit relative to the equator.
 - ▶ **Equatorial** or **Polar** paths.
 - ▶ Degrees N
(90°for polar orbit, > 90°**retrograde**).
- ▶ **Altitude:** Distance from Earth, affects speed and coverage.
 - ▶ Speed depends on altitude; lower orbits move faster.
 - ▶ **LEO:** Low Earth Orbit, close, high-resolution.
 - ▶ **GEO:** Geosynchronous Earth Orbit, far, stationary view.





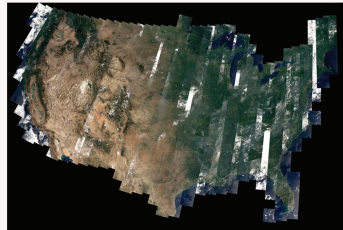
Low Earth Orbit (LEO)

- ▶ **LEO:** Altitude of up to 2,000 km.
 - ▶ Closer to Earth's surface.
 - ▶ Provides high-resolution imagery.
- ▶ Common for scientific and weather satellites.
 - ▶ **E.g.** Landsat satellites.



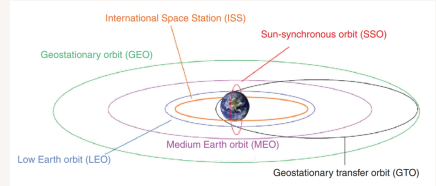
Sun-Synchronous Orbit (SSO)

- ▶ **SSO:** Special LEO orbit
 - ▶ Inclined at approximately 98° .
 - ▶ 'Twists' (precesses) $365.25/360$ degrees per day (orbit faces sun)
 - ▶ Hits same latitude at the same solar time each day.
 - ▶ Longitudinal shift of 1 degree per day - but swaths are not longitudinal!.
 - ▶ Daily spin leads to diagonal swath pattern.
- ▶ Ideal for imaging systems reliant on sunlight.
 - ▶ Reduces lighting differences between passes.
- ▶ Common for environmental and land observation satellites.



Medium Earth Orbit (MEO)

- ▶ **MEO:** Altitudes between 2,000 km and 35,786 km.
- ▶ Satellites orbit at approximately 20,200 km altitude.
- ▶ Used for **navigation systems** like GPS, GLONASS, and Galileo.
- ▶ Satellites have a **12-hour orbital period** (two orbits per Earth day).



Geostationary Orbit (GEO)

- ▶ **GEO:** Altitude approximately 35,786 km.
- ▶ Satellite stays fixed over a specific point on Earth's equator.
- ▶ Ideal for continuous monitoring of weather patterns and communication networks.
- ▶ **Wide-area coverage**, but lower spatial resolution than lower orbits.

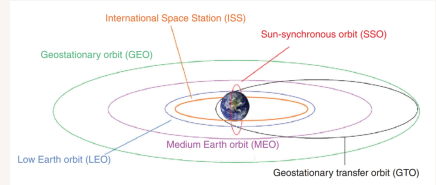
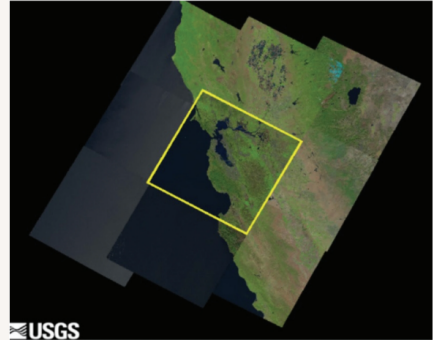


Image Footprint in Remote Sensing

- ▶ **Footprint:** Area viewed by a satellite's sensor during a single pass.
- ▶ Defines the satellite's **swath width** (total width of the viewed area).
- ▶ Satellite footprints vary in size based on orbit altitude and sensor resolution.
 - ▶ **LEO satellites:** Small footprint, high resolution.
 - ▶ **GEO satellites:** Large footprint, lower resolution.



Satellite Constellations

- ▶ **Satellite constellation:** Group of satellites working together to provide continuous coverage.
- ▶ Example: GPS system relies on multiple satellites working in sync.
- ▶ Enables **better temporal and spatial coverage**.
- ▶ Constellations can involve different types of satellites sharing similar orbits.



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Unmanned Aerial Systems (UAS)

- ▶ **UAS:** Remotely piloted or autonomous aerial systems.
- ▶ Long history in **military applications**; now used in science, agriculture, and more.
- ▶ Provide **flexibility** for missions in dangerous or inaccessible environments.
- ▶ Widely used for detailed aerial imagery and monitoring.



UAS Sensors

- ▶ UAS can carry various sensors:
 - ▶ **Visible light sensors:** Capture blue, green, and near-infrared imagery.
 - ▶ **Thermal infrared sensors:** Detect heat signatures.
 - ▶ **Multispectral sensors:** Capture a wide range of wavelengths for specialized analysis.
- ▶ Compact and lightweight, allowing flexibility in sensor choice.



UAS Flight Planning and Operations

- ▶ **Autonomous flight planning:** UAS flight paths can be pre-programmed.
- ▶ Uses **geofencing**: Virtual boundaries to keep UAS within designated areas.
- ▶ Systems monitor battery life to return UAS to its base before power runs out.



Applications of UAS

- ▶ **Agriculture:** Inspect and monitor fields and soil conditions.
- ▶ **Forestry:** Detect forest health and monitor growth.
- ▶ **Emergency response:** Assist in disaster monitoring (fires, floods).
- ▶ **Infrastructure monitoring:** Check pipelines, powerlines, and towers.

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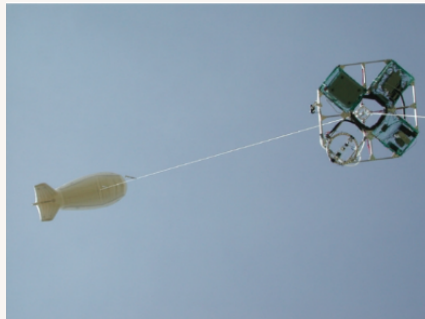
Tethered Balloons

- ▶ **Tethered balloons:** Platforms for remote sensing that remain stationary.
- ▶ Long history in military applications, now used for **archaeology** and **wildlife monitoring**.
- ▶ **Low cost** and can carry heavy sensor equipment.



Applications of Tethered Balloons

- ▶ **Wildlife monitoring:** Continuous observation of wildlife in remote areas.
- ▶ **Archaeology:** Track excavation progress with sequential photography.
- ▶ **Surveillance:** Used for border monitoring and environmental observation.



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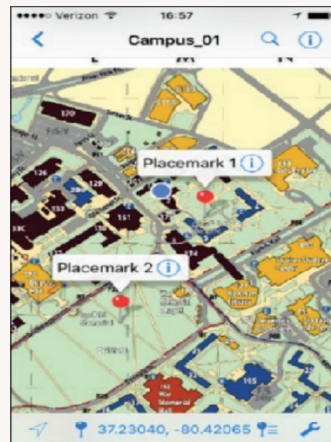
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Mobile Collection of Field Data

Summary

Mobile Collection of Field Data

- ▶ Mobile devices (phones, tablets) allow **real-time data collection**.
- ▶ Can integrate field data with **GIS layers** for instant analysis.
- ▶ Examples: Accuracy assessment, training data for image classification.



Mobile Data Collection Apps

- ▶ **Collector for ArcGIS:** Collect and update GIS data in the field.
- ▶ **Nature's Notebook:** Record phenological data on-site.
- ▶ **Avenza Maps:** Use digital maps to collect spatial data.
- ▶ **Fulcrum App:** Mobile platform for environmental surveys and field data collection.



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Summary of RS Platforms

- ▶ **Fixed-Wing Aircraft:** Long endurance, heavy payload.
- ▶ **Helicopters:** Hover, confined spaces.
- ▶ **Satellites:** Global coverage, long-term monitoring.
- ▶ **UAS:** Flexible, detailed local monitoring.
- ▶ **Tethered Balloons:** Long-term stationary observations.

